

X05 ENHANCED MATERIAL RESTRICTIONS | O (MAX : 2 PT)

Intent : Minimize the exposure to certain chemicals by limiting their presence in products.

Summary : This WELL feature requires restricting chemicals found in products commonly installed in buildings.

Issue : The materials industry has been able to develop and adapt products to satisfy the needs of the market. However, some newly introduced chemicals that pose to be advantageous from a performance or cost perspective, may be associated with negative health effects. Populations that prove to be the most vulnerable to these chemicals are unborn and young children, as well as pregnant women.^{1,2} Chemical classes with compounds suspected or proven to pose health concerns include orthophthalates (common plasticizers), halogenated flame retardants (HFR)^{3,4} and other per-fluorinated compounds (PFCs) and heavy metals, in addition to formaldehyde.⁴ Common pathways of exposure to these compounds are through inhalation, skin contact or swallowing of dusts, soils or larger particles. Overall, there is a great research gap in assessing the safety of these and many other chemicals,⁵ driven both by their ubiquity in the environment, as well as their presence in the human body. For instance, Per- and polyfluoroalkyl substances (PFAS) such as perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) are persisting PFCs in natural environments (including sources of drinking water) and found in humans.⁶ Metabolites of orthophthalates are partially metabolized by the body and commonly found in urine.⁷

Solutions : Selecting products that are devoid of or have reduced amount of chemicals associated with health concerns may help to prevent exposure to these compounds. Some organizations have published regulations, guidelines or certifications that restrict or minimize the presence of certain chemicals from these classes.

PART 1 SELECT COMPLIANT INTERIOR FURNISHINGS (MAX : 1 PT)

For All Spaces:

1: Furniture, millwork and fixtures

At least 50% by cost of newly installed furniture, millwork and fixtures (minimum 10 distinct products), as defined in Appendix X1, meet one of the following requirements:

- a. Textiles (i.e., fabrics including upholstery) and plastics in products contain 100 ppm (0.01%) by weight or less of the below compounds and chemical classes, unless higher amounts are mandated by local codes. For assessing compliance of a product, all pieces of each of the two material categories (textiles, plastics) are grouped together and each material category is assessed independently against the 100 ppm threshold:
 1. Halogenated flame retardants (HFR).
 2. Per- and polyfluoroalkyl substances (PFAS).
 3. Lead.
 4. Cadmium.
 5. Mercury.
- b. Do not contain textiles and plastic.

2: Electrical and electronic products

All newly installed electrical and electronic products, as specified in Appendix X1, meet the following requirement:

- a. Products are compliant with RoHS restrictions.

PART 2 SELECT COMPLIANT ARCHITECTURAL AND INTERIOR PRODUCTS (MAX : 1 PT)

For All Spaces:

At least 50% by cost of newly installed products under the classes listed below, as defined by [Appendix X1](#) (minimum 10 distinct products), meet the following requirements, unless higher amounts are mandated by local code:

- a. Flooring products contain 100 ppm (0.01%) by weight or less of the following:
 1. Halogenated flame retardants (HFR).
 2. Per- and polyfluoroalkyl substances (PFAS).
 3. Orthophthalates.
- b. Insulation products, including thermal and acoustic insulation in walls, ceilings, ducts, tubes and pipes, contain 100 ppm (0.01%) by weight or less of halogenated flame retardants (HFR).
- c. Ceiling and wall finishes and demountable wall partitions contain 100 ppm (0.01%) by weight or less of the following:
 1. Halogenated flame retardants (HFR).
 2. Orthophthalates.
- d. Pipes and fittings intended for drinking water distribution and delivery contains 100 ppm (0.01%) by weight or less of orthophthalates.

REFERENCES

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